Lesson 6.9 — Logarithmic Scales

Big idea: Many natural quantities span many orders of magnitude (sound intensity, earthquake energy, acidity). A log scale spreads them out so each step is a constant multiple of the previous.

Quick review

pH: $\mathrm{pH} = -\log_{10}([H^+])$. Lower pH = more acidic. A change of 1 unit = 10× concentration.
Decibels (dB): $L = 10\log_{10}(I/I_0)$, where I_0 is the threshold of hearing. +10 dB = 10× the intensity.
Richter scale: magnitude M, where amplitude scales as 10^M. +1 magnitude = 10× amplitude, ≈ 32× energy.
Common pattern: a log scale linearizes exponential phenomena.

Worked example

Worked example. Sound 1 is 80 dB; sound 2 is 100 dB. How many times more intense is sound 2?

100 – 80 = 20 dB difference ⇒ intensity ratio $10^{20/10} = 100\times$.

Warm-ups

1. Lemon juice has pH 2; water has pH 7. How many times more acidic is lemon juice (in $[H^+]$)?
2. Sound intensity rises 30 dB. How many times more intense?
3. Earthquake A is magnitude 5, B is 6. How much more amplitude does B have than A?

Practice

1. If $[H^+] = 1 \times 10^{-4}$, find pH.
2. A solution has pH 8. Find its $[H^+]$.
3. Whisper ≈ 30 dB. Conversation ≈ 60 dB. Ratio of intensities?

Challenge

1. Earthquake X has magnitude 7.2; Y has magnitude 5.5. How much more amplitude (and roughly more energy, using the 32× per unit estimate) does X have?
2. If sound A is 1000× more intense than the threshold of hearing, what is its dB level?

Answer key — Lesson 6.9

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $10^5 = 100{,}000\times$ more acidic.
2. $10^3 = 1{,}000\times$ more intense.
3. $10\times$ the amplitude ($\approx 32\times$ energy).

Practice

1. pH = 4.
2. 10^{-8} mol/L.
3. $10^3 = 1{,}000\times$ more intense.

Challenge

1. Difference 1.7. Amplitude: $10^{1.7} \approx 50\times$ more. Energy: $32^{1.7} \approx 322\times$ more.
2. $L = 10 \log_{10}(1000) = 10 \cdot 3 = 30$ dB.